

Multiple Choice

1. **Answer:** A

Options: A) Am; B) As; C) Ab; D) Au; E) Fe

Note: Correct option: A - Am

2. **Answer:** D

Options: A) 2.11×10^{-8} ; B) 2.73×10^{-8} ; C) 1.08×10^{-8} ; D) 3.92×10^{-8} ; E) 4.77×10^{-8}

Note: Correct option : D - 3.92×10^{-8}

2. **Answer:** C

Options: A) $1\text{ s} < 2\text{ p} < 2\text{ s} < 3\text{ p} < 3\text{ s} < 3\text{ d}$; B) $2\text{ s} < 2\text{ p} < 1\text{ s} < 3\text{ s} < 3\text{ p} < 3\text{ d}$;

C) $1\text{ s} < 2\text{ s} = 2\text{ p} < 3\text{ s} = 3\text{ p} = 3\text{ d} < 4\text{ s} = 4\text{ p} = 4\text{ d} = 4\text{ f} < 5\text{ s} = 5\text{ p} = 5\text{ d} = 5\text{ f} = 5\text{ g}$; D) $2\text{ s} = 2\text{ p} < 1\text{ s} = 3\text{ p} = 3\text{ d}$; E) $1\text{ s} < 3\text{ s} < 2\text{ s} < 5\text{ s} < 4\text{ s}$

Note: Correct option: C - $1\text{ s} < 2\text{ s} = 2\text{ p} < 3\text{ s} = 3\text{ p} = 3\text{ d} < 4\text{ s} = 4\text{ p} = 4\text{ d} = 4\text{ f} < 5\text{ s} = 5\text{ p} = 5\text{ d} = 5\text{ f} = 5\text{ g}$

3. **Answer:** C

Options: A) $1.25 \times 10^3\text{ K}$; B) $1.20 \times 10^3\text{ K}$; C) $1.41 \times 10^3\text{ K}$; D) $1.10 \times 10^3\text{ K}$; E) $1.65 \times 10^3\text{ K}$

Note: Correct option: C - $1.41 \times 10^3\text{ K}$

4. **Answer:** B

Options: A) Al; B) Tl; C) C; D) B; E) Si

Note: Correct option: B - Tl

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'covalent network bonding'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** CH 3 Cl. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Neutrons. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key